University College London, 2026–2027
Econ 0107 – Macroeconomics I – Franck Portier

Problem Set: Dynamic Optimization
Chapter 4 in Azzimonti, Krusell, McKay, Mukoyama and others's textbook

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Problem 1 – Two-period consumption-saving model

A representative agent lives for two periods ($t = 0, 1$). She receives wage income $w_0 > 0$ in period 0 and $w_1 > 0$ in period 1, and can save or borrow at the constant net interest rate $r > 0$. Let a_1 denote savings carried into period 1, with the borrowing constraint $a_1 \geq 0$ and the terminal condition $a_2 = 0$. Preferences are $u(c_0) + \beta u(c_1)$ with $u(c) = \log c$ and $0 < \beta < 1$.

1. Substitute the two period budget constraints to obtain the *lifetime budget constraint* and write the Lagrangian associated to the constrained maximisation problem.

2. Derive the Kuhn-Tucker first-order conditions. Assuming an interior solution ($a_1 > 0$), derive the Euler equation.

3. Suppose $\beta(1+r) = 1$. Show that optimal consumption is constant across periods, $c_0^* = c_1^*$, and find its value. Verify that the interior-solution assumption is satisfied when $w_0 > w_1/(1+r)$.

4. Now suppose $\beta(1+r) < 1$ and $w_1 > \beta(1+r)w_0$. Show that the borrowing constraint binds at the optimum, find (c_0^*, c_1^*), and compute the shadow price λ of the constraint.

Solutions – Problem 1

*Answer to **1**.* The budget constraints are $c_0 = w_0 - a_1$ and $c_1 = w_1 + (1+r)a_1$. Solving the second for a_1 and substituting into the first gives the *lifetime budget constraint*

$$c_0 + \frac{c_1}{1+r} = w_0 + \frac{w_1}{1+r} \equiv W.$$

The Lagrangian, treating c_0 and c_1 as the choice variables, is

$$\mathcal{L} = \log c_0 + \beta \log c_1 + \mu\left(W - c_0 - \frac{c_1}{1+r}\right) + \lambda(w_0 - c_0),$$

where $\mu \geq 0$ is the multiplier on the lifetime budget constraint and $\lambda \geq 0$ is the Kuhn-Tucker multiplier on the borrowing constraint $a_1 = w_0 - c_0 \geq 0$. (The Inada condition $u'(0) = \infty$ ensures $c_t > 0$, so non-negativity of consumption is never binding.)

*Answer to **2**.* First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial c_0}: \quad \frac{1}{c_0} = \mu + \lambda,$$
$$\frac{\partial \mathcal{L}}{\partial c_1}: \quad \frac{\beta}{c_1} = \frac{\mu}{1+r}.$$

Kuhn-Tucker: $\lambda(w_0 - c_0) = 0$, $w_0 - c_0 \geq 0$, $\lambda \geq 0$.

At an interior solution $w_0 - c_0 > 0$, so $\lambda = 0$ and $\mu = 1/c_0$. Substituting into the second condition gives the *Euler equation*

$$\frac{1}{c_0} = \beta(1+r)\,\frac{1}{c_1}, \qquad \text{i.e.,} \qquad u'(c_0) = \beta(1+r)\,u'(c_1).$$

This equates the marginal cost of saving one unit today (foregone utility $1/c_0$) with the discounted marginal benefit ($(1+r)$ units of consumption next period, worth β/c_1 in utility).

*Answer to **3**.* When $\beta(1+r) = 1$ the Euler equation gives $c_1/c_0 = 1$, i.e., $c_0 = c_1 = c^*$. Substituting into the lifetime budget constraint:

$$c^*\left(1 + \frac{1}{1+r}\right) = W \implies c^* = \frac{(1+r)W}{2+r} = \frac{(1+r)\big(w_0 + w_1/(1+r)\big)}{2+r}.$$

Optimal savings are $a_1^* = w_0 - c^*$. Substituting:

$$a_1^* = w_0 - \frac{(1+r)w_0 + w_1}{2+r} = \frac{w_0(2+r) - (1+r)w_0 - w_1}{2+r} = \frac{w_0 - w_1}{2+r}.$$

Since $w_0 > w_1/(1+r)$ implies $w_0(1+r) > w_1$, and $(2+r) > 0$, we have $a_1^* > 0$ provided $w_0 > w_1$. More generally, $a_1^* \geq 0$ iff $w_0 \geq w_1$, confirming the interior assumption.

*Answer to **4**.* Suppose $\beta(1+r) < 1$ (the agent is impatient) and $w_1 > \beta(1+r)w_0$ (income rises sharply). The unconstrained optimal $a_1^u < 0$ (the agent wishes to borrow); since borrowing is forbidden, the constraint binds: $a_1^* = 0$. Then $c_0^* = w_0$ and $c_1^* = w_1$. The shadow price follows from the first-order condition for c_0:

$$\frac{1}{c_0^*} = \mu + \lambda \quad \text{and} \quad \frac{\beta}{c_1^*} = \frac{\mu}{1+r},$$

so $\mu = \beta(1+r)/w_1$ and $\lambda = 1/w_0 - \beta(1+r)/w_1 > 0$ (verified under the stated assumption). The shadow price measures the utility gain from relaxing the borrowing limit by one unit.

Problem 2 – Backward induction in the finite NGM ($T = 2$)

Consider the social planner's problem in the neoclassical growth model (NGM) with $T = 2$ (periods $t = 0, 1, 2$, terminal condition $k_3 = 0$). Utility is logarithmic $u(c) = \log c$, the production function is Cobb-Douglas $f(k) = Ak^\alpha$ ($0 < \alpha < 1$, $A > 0$), and there is full depreciation ($\delta = 1$). The initial capital stock $k_0 > 0$ is given.

1. Write the planner's problem and derive the Euler equations for $t = 0$ and $t = 1$.

2. Use backward induction, starting from $t = 1$, to obtain explicit expressions for k_2 as a function of k_1, and for k_1 as a function of k_0.

3. Express the optimal saving rates $s_t \equiv k_{t+1}/(Ak_t^\alpha)$ for $t = 0$ and $t = 1$. Verify they match the general formula from the text: $s_t = \alpha\beta \dfrac{1 - (\alpha\beta)^{T-t}}{1 - (\alpha\beta)^{T-t+1}}$.

4. What do s_0 and s_1 converge to as $T \to \infty$? Interpret the result.

Solutions – Problem 2

*Answer to **1**.* The planner solves

$$\max_{\{k_1, k_2\}} \log(Ak_0^\alpha - k_1) + \beta \log(Ak_1^\alpha - k_2) + \beta^2 \log(Ak_2^\alpha),$$

using the terminal condition $k_3 = 0$ (which is optimal by the Inada condition on u). First-order conditions with respect to k_1 and k_2:

$$\begin{aligned} [k_1]: &\quad \frac{-1}{Ak_0^\alpha - k_1} + \frac{\beta\alpha A k_1^{\alpha-1}}{Ak_1^\alpha - k_2} = 0, \\ [k_2]: &\quad \frac{-1}{Ak_1^\alpha - k_2} + \frac{\beta^2\alpha A k_2^{\alpha-1}}{\beta\, Ak_2^\alpha} = 0. \end{aligned}$$

Simplifying, the two Euler equations are

$$\begin{aligned} \frac{1}{c_0} &= \beta \frac{\alpha A k_1^{\alpha-1}}{c_1}, \qquad (t = 0) \\ \frac{1}{c_1} &= \beta \frac{\alpha A k_2^{\alpha-1}}{c_2}. \qquad (t = 1) \end{aligned}$$

*Answer to **2**. Step 1 ($t = 1$).* Since $k_3 = 0$ the resource constraint gives $c_2 = Ak_2^\alpha$, so the $t = 1$ Euler equation becomes

$$\frac{1}{Ak_1^\alpha - k_2} = \frac{\beta\alpha}{k_2} \implies k_2\big[1 + \alpha\beta\big] = \alpha\beta A k_1^\alpha \implies \boxed{k_2 = \frac{\alpha\beta}{1+\alpha\beta} Ak_1^\alpha.}$$

Step 2 ($t = 0$). Substitute k_2 into the $t = 0$ Euler equation. We have $c_1 = Ak_1^\alpha - k_2 = Ak_1^\alpha/(1+\alpha\beta)$, so

$$\frac{1}{Ak_0^\alpha - k_1} = \frac{\beta\alpha A k_1^{\alpha-1}(1+\alpha\beta)}{Ak_1^\alpha} = \frac{\beta\alpha(1+\alpha\beta)}{k_1}.$$

Solving for k_1:

$$\boxed{k_1 = \frac{\alpha\beta(1+\alpha\beta)}{1+\alpha\beta+(\alpha\beta)^2}\, Ak_0^\alpha.}$$

*Answer to **3**.* Reading off the saving rates:

$$s_1 = \frac{k_2}{Ak_1^\alpha} = \frac{\alpha\beta}{1+\alpha\beta}, \qquad s_0 = \frac{k_1}{Ak_0^\alpha} = \frac{\alpha\beta(1+\alpha\beta)}{1+\alpha\beta+(\alpha\beta)^2}.$$

The general formula is $s_t = \alpha\beta\,\frac{1-(\alpha\beta)^{T-t}}{1-(\alpha\beta)^{T-t+1}}$. For $T = 2$:

$$s_1 = \alpha\beta\,\frac{1-(\alpha\beta)^1}{1-(\alpha\beta)^2} = \alpha\beta\,\frac{1-\alpha\beta}{(1-\alpha\beta)(1+\alpha\beta)} = \frac{\alpha\beta}{1+\alpha\beta}. \checkmark$$
$$s_0 = \alpha\beta\,\frac{1-(\alpha\beta)^2}{1-(\alpha\beta)^3} = \alpha\beta\,\frac{(1-\alpha\beta)(1+\alpha\beta)}{(1-\alpha\beta)(1+\alpha\beta+(\alpha\beta)^2)} = \frac{\alpha\beta(1+\alpha\beta)}{1+\alpha\beta+(\alpha\beta)^2}. \checkmark$$

*Answer to **4**.* Since $\alpha\beta < 1$, as $T \to \infty$ both $(\alpha\beta)^{T-t} \to 0$ and $(\alpha\beta)^{T-t+1} \to 0$ for any fixed t, so

$$s_t \;\longrightarrow\; \alpha\beta\,\frac{1-0}{1-0} = \alpha\beta.$$

In the infinite-horizon economy the optimal saving rate is *constant* at $\alpha\beta$, independent of time and the level of capital. This mirrors the Solow model's constant saving rate, but here it emerges endogenously from optimisation (the log/Cobb-Douglas/full-depreciation case is the unique closed-form example with these properties). In the finite-horizon economy, agents reduce their saving rate as the terminal period approaches because they will not benefit from capital accumulated after the last period.

Problem 3 – CRRA utility and the elasticity of intertemporal substitution

Consider a representative agent with CRRA utility $u(c) = \dfrac{c^{1-\sigma}-1}{1-\sigma}$, $\sigma > 0$, $\sigma \neq 1$, facing a linear production technology $f_t(k_t) = R_t k_t$ (with time-varying gross return $R_t > 0$). The agent maximises $\sum_{t=0}^{\infty} \beta^t u(c_t)$ subject to the sequence of budget constraints.

1. Show that the Euler equation between periods t and $t+1$ is $u'(c_t) = \beta R_{t+1}\, u'(c_{t+1})$.

2. By iterating the Euler equation, show that $\dfrac{c_{t+s}}{c_t} = \left(\beta^s R_{t,t+s}\right)^{1/\sigma}$ where $R_{t,t+s} \equiv R_{t+1}R_{t+2}\cdots R_{t+s}$.

3. Compute the elasticity of intertemporal substitution (EIS), defined as

$$\text{EIS} \equiv \frac{\partial \log(c_{t+s}/c_t)}{\partial \log R_{t,t+s}}.$$

4. Interpret the cases $\sigma < 1$, $\sigma = 1$, and $\sigma > 1$ in terms of income and substitution effects when $R_{t,t+s}$ rises by 1%.

Solutions – Problem 3

*Answer to **1**.* The per-period return on capital is R_t and the Bellman equation implies $u'(c_t) = \beta R_{t+1} u'(c_{t+1})$. For CRRA, $u'(c) = c^{-\sigma}$, so

$$c_t^{-\sigma} = \beta R_{t+1}\, c_{t+1}^{-\sigma}.$$

*Answer to **2**.* Iterating s steps: $c_t^{-\sigma} = \beta^s c_{t+s}^{-\sigma} \cdot R_{t,t+s}$, where $R_{t,t+s} = \prod_{j=1}^{s} R_{t+j}$. Rearranging:

$$\left(\frac{c_{t+s}}{c_t}\right)^{-\sigma} = \beta^s R_{t,t+s} \implies \frac{c_{t+s}}{c_t} = \left(\beta^s R_{t,t+s}\right)^{1/\sigma}.$$

*Answer to **3**.* Taking logs: $\log(c_{t+s}/c_t) = \frac{1}{\sigma}\big[s\log\beta + \log R_{t,t+s}\big]$. Therefore

$$\text{EIS} = \frac{\partial \log(c_{t+s}/c_t)}{\partial \log R_{t,t+s}} = \frac{1}{\sigma}.$$

The EIS is constant (independent of the consumption level), which is why the CRRA function is also called CEIS (Constant Elasticity of Intertemporal Substitution).

*Answer to **4**.* A 1 % increase in $R_{t,t+s}$ raises c_{t+s}/c_t by $1/\sigma$ %.

- **Substitution effect:** a higher return makes future consumption cheaper relative to current consumption, inducing the agent to save more and shift consumption towards the future.
- **Income effect:** the higher return raises wealth, which the agent spreads over all periods (including period t), reducing the incentive to save.
- $\sigma < 1$: EIS> 1. The substitution effect dominates. A rise in $R_{t,t+s}$ raises c_{t+s}/c_t, i.e. future consumption rises relative to current consumption; the agent saves more.
- $\sigma = 1$ (log utility): EIS$= 1$. Income and substitution effects exactly cancel. The consumption ratio c_{t+s}/c_t is proportional to $R_{t,t+s}$, and the expenditure share on current consumption is constant.
- $\sigma > 1$: EIS< 1. The income effect dominates. A rise in $R_{t,t+s}$ still raises the ratio c_{t+s}/c_t (the substitution effect is never reversed), but by less than 1 % per 1 % increase in the return, so saving rises less than proportionately.

Problem 4 – No Ponzi Game condition and the Transversality Condition

Consider an infinitely-lived agent facing the budget constraint $c_t + a_{t+1} = (1+r)a_t$ for all $t \geq 0$, with initial wealth $a_0 > 0$ and constant interest rate $r > 0$. The agent maximises $\sum_{t=0}^{\infty} \beta^t u(c_t)$ with u strictly increasing.

1. After T substitutions of successive budget constraints, show that

$$\sum_{t=0}^{T} \frac{c_t}{(1+r)^t} = a_0(1+r) - \frac{a_{T+1}}{(1+r)^T}.$$

2. State the no-Ponzi-game (nPG) condition and derive the infinite-horizon lifetime budget constraint (LBC) by taking $T \to \infty$.

3. Define the transversality condition (TVC). Using the assumption that u is strictly increasing and assets yield no direct utility, argue that an optimising agent chooses $\lim_{T\to\infty} a_{T+1}/(1+r)^T = 0$, so the LBC holds with equality.

4. Explain clearly the conceptual difference between the nPG condition and the TVC. Which one is externally imposed and which one is an optimality condition?

Solutions – Problem 4

*Answer to **1**.* From the period-0 constraint: $a_1 = (1+r)a_0 - c_0$. Substituting into the period-1 constraint and rearranging: $a_2 = (1+r)^2 a_0 - (1+r)c_0 - c_1$. Iterating T times gives

$$a_{T+1} = (1+r)^{T+1} a_0 - \sum_{t=0}^{T} (1+r)^{T-t} c_t.$$

Dividing by $(1+r)^T$:

$$\frac{a_{T+1}}{(1+r)^T} = (1+r)a_0 - \sum_{t=0}^{T} \frac{c_t}{(1+r)^t},$$

which rearranges to the stated identity.

*Answer to **2**.* The **no-Ponzi-game condition** requires

$$\lim_{T\to\infty} \frac{a_{T+1}}{(1+r)^T} \geq 0.$$

Intuitively, the present value of the agent's net debt cannot be positive at infinity; otherwise, she would be rolling over debt forever without ever repaying it, which no lender would accept.

Taking $T \to \infty$ in Question 1 and using the nPG condition:

$$\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = (1+r)a_0 - \lim_{T\to\infty} \frac{a_{T+1}}{(1+r)^T} \leq (1+r)a_0.$$

This is the **lifetime budget constraint (LBC)**.

*Answer to **3**.* The **transversality condition** is

$$\lim_{T\to\infty} \frac{a_{T+1}}{(1+r)^T} \leq 0.$$

To see why an optimiser must satisfy it: suppose the limit were strictly positive, say equal to $\varepsilon > 0$. Then the agent is leaving positive present-value wealth unconsumed as $T \to \infty$. She could instead raise c_0 by $\varepsilon \cdot (1+r)$ while keeping the plan feasible (adjusting a_{T+1} downward), strictly increasing utility because u is strictly increasing. This contradicts optimality. Hence $\lim_{T\to\infty} a_{T+1}/(1+r)^T \leq 0$, and combined with the nPG condition the LBC holds with equality.

*Answer to **4**.*

- The **nPG condition** ($\lim a_{T+1}/(1+r)^T \geq 0$) is an *externally imposed institutional constraint*. It prevents the agent from running a "Ponzi scheme" – borrowing without bound and never repaying. Markets (lenders) impose it because no rational lender would finance such a scheme. It rules out *too much debt*.
- The **TVC** ($\lim a_{T+1}/(1+r)^T \leq 0$) is a *self-imposed optimality condition*. It follows from the agent's own objectives: it is never optimal to leave positive present-value wealth unconsumed, because assets yield no direct utility. It rules out *excessive saving*.

Together, nPG + TVC $\Rightarrow \lim a_{T+1}/(1+r)^T = 0$, i.e. the LBC holds with equality.

Problem 5 – Infinite-horizon NGM: guess and verify

Consider the infinite-horizon social planner's problem in the NGM with $u(c) = \log c$, $f(k) = Ak^\alpha$, and full depreciation $\delta = 1$.

1. Write the Euler equation and show that the functional Euler equation (FEE) for the policy function $g(k) = k'$ is

$$\frac{1}{Ak^\alpha - g(k)} = \frac{\beta\alpha A\, g(k)^{\alpha-1}}{A\, g(k)^\alpha - g(g(k))} \quad \forall k.$$

2. Guess that $g(k) = sAk^\alpha$ for some constant s (linear saving rule). Substitute into the FEE and solve for s.

3. Find the steady-state capital stock $\bar{k}$ implied by this policy function.

4. Verify that the transversality condition holds for the solution found in Question 2.

Solutions – Problem 5

*Answer to **1**.* The planner maximises $\sum_{t=0}^{\infty} \beta^t \log(Ak_t^\alpha - k_{t+1})$. The Euler equation is $u'(c_t) = \beta u'(c_{t+1})\big[f'(k_{t+1}) + 1 - \delta\big]$. With $u'(c) = 1/c$, $f'(k) = \alpha Ak^{\alpha-1}$, and $\delta = 1$:

$$\frac{1}{c_t} = \frac{\beta\alpha Ak_{t+1}^{\alpha-1}}{c_{t+1}}.$$

Writing $c_t = Ak_t^\alpha - g(k_t)$ and $c_{t+1} = Ag(k_t)^\alpha - g(g(k_t))$, the functional Euler equation follows immediately.

*Answer to **2**.* Substitute $g(k) = sAk^\alpha$ and $g(g(k)) = sA(sAk^\alpha)^\alpha = s^{1+\alpha}A^{1+\alpha}k^{\alpha^2}$:

$$\frac{1}{(1-s)Ak^\alpha} = \frac{\beta\alpha A(sAk^\alpha)^{\alpha-1}}{A(sAk^\alpha)^\alpha - s^{1+\alpha}A^{1+\alpha}k^{\alpha^2}}.$$

The right-hand side simplifies: $A(sAk^\alpha)^\alpha = s^\alpha A^{1+\alpha}k^{\alpha^2}$, so $A(sAk^\alpha)^\alpha - g(g(k)) = (s^\alpha - s^{1+\alpha})A^{1+\alpha}k^{\alpha^2} = (1-s)s^\alpha A^{1+\alpha}k^{\alpha^2}$. Thus the right-hand side becomes $\beta\alpha(sAk^\alpha)^{\alpha-1}/((1-s)s^{\alpha-1}Ak^{\alpha^2/\alpha}) = \beta\alpha/((1-s)sAk^\alpha)$. Equating to the left-hand side:

$$\frac{1}{(1-s)Ak^\alpha} = \frac{\beta\alpha}{(1-s)sAk^\alpha} \implies s = \alpha\beta.$$

The optimal policy is $g(k) = \alpha\beta Ak^\alpha$, i.e. a constant saving rate $s^* = \alpha\beta$.

*Answer to **3**.* At the steady state $g(\bar{k}) = \bar{k}$, so $\alpha\beta A\bar{k}^\alpha = \bar{k}$, giving

$$\bar{k} = (\alpha\beta A)^{1/(1-\alpha)}.$$

This matches equation (4.16) of the text with $\delta = 1$: $1 = \beta(f'(\bar{k}) + 1 - 1) = \beta\alpha A\bar{k}^{\alpha-1}$.

*Answer to **4**.* The TVC requires $\lim_{t\to\infty}\beta^t u'(c_t)f'(k_t)k_t = 0$. With $u'(c) = 1/c$, $c_t = (1-\alpha\beta)Ak_t^\alpha$ (since $k_{t+1} = \alpha\beta Ak_t^\alpha$ and $c_t = Ak_t^\alpha - k_{t+1}$), and $f'(k_t) = \alpha Ak_t^{\alpha-1}$:

$$\beta^t u'(c_t)f'(k_t)k_t = \beta^t \frac{\alpha Ak_t^{\alpha-1}}{(1-\alpha\beta)Ak_t^\alpha}\cdot k_t = \frac{\alpha\beta^t}{1-\alpha\beta}.$$

Since $\beta < 1$, this tends to zero as $t \to \infty$. The TVC is satisfied.

Problem 6 – Value function iteration: the Bellman equation

Use the same parametric NGM as in Problem 5 ($u(c) = \log c$, $f(k) = Ak^\alpha$, $\delta = 1$, infinite horizon).

1. Write the Bellman equation for this problem.

2. Starting from the initial guess $V_0(k) = 0$ for all k, compute $V_1(k)$ and $V_2(k)$ analytically. Identify the implied policy function at each iteration.

3. Guess that the value function takes the form $V(k) = a + b\log k$ and substitute into the Bellman equation. Derive the system of equations that a and b must satisfy and solve for b.

4. Show that the policy function obtained from the fixed-point of the Bellman equation matches the solution found in Problem 5.

Solutions – Problem 6

*Answer to **1**.* The Bellman equation is

$$V(k) = \max_{k'\geq 0}\{\log(Ak^\alpha - k') + \beta V(k')\}.$$

*Answer to **2**. Iteration 1.* $V_1(k) = \max_{k'\geq 0}\{\log(Ak^\alpha - k')\}$. The objective is decreasing in k', so $k' = 0$ is optimal. Hence $V_1(k) = \log(Ak^\alpha) = \log A + \alpha\log k$.

Iteration 2. $V_2(k) = \max_{k'\geq 0}\{\log(Ak^\alpha - k') + \beta(\log A + \alpha\log k')\}$. The first-order condition is

$$\frac{1}{Ak^\alpha - k'} = \frac{\beta\alpha}{k'} \implies k' = \frac{\alpha\beta}{1+\alpha\beta}Ak^\alpha.$$

Substituting back:

$$\begin{aligned} V_2(k) &= \log\left(\frac{Ak^\alpha}{1+\alpha\beta}\right) + \beta\left[\log A + \alpha\log\left(\frac{\alpha\beta Ak^\alpha}{1+\alpha\beta}\right)\right] \\ &= \underbrace{\log A + \beta\log A + \alpha\beta\log\frac{\alpha\beta A}{1+\alpha\beta} - \log(1+\alpha\beta) - \alpha\beta\log(1+\alpha\beta)}_{=a_2} + \underbrace{\alpha(1+\alpha\beta)}_{=b_2}\log k. \end{aligned}$$

Both V_1 and V_2 are log-linear in k. The policy at iteration 2 is $k' = \alpha\beta Ak^\alpha/(1+\alpha\beta)$, which corresponds to the $T = 1$ saving rule from Problem 2.

Answer to 3. Substituting $V(k) = a + b\log k$ into the Bellman equation, the first-order condition gives

$$k' = \frac{\beta b}{1+\beta b}\, Ak^{\alpha}.$$

Substituting back and collecting terms in $\log k$:

$$a + b\log k = \underbrace{\log A - \log(1+\beta b) + \beta a + \beta b \log\frac{\beta b A}{1+\beta b}}_{=\text{constant}} + \underbrace{\alpha(1+\beta b)}\log k.$$

Matching coefficients of $\log k$: $b = \alpha(1+\beta b)$, so $b(1-\alpha\beta) = \alpha$, giving

$$\boxed{b = \frac{\alpha}{1-\alpha\beta}.}$$

Matching the constant terms determines a (whose exact form is given in the text); existence and uniqueness follow from $\beta < 1$.

Answer to 4. Using $b = \alpha/(1-\alpha\beta)$, we have $1 + \beta b = 1/(1-\alpha\beta)$, so

$$k' = \frac{\beta b}{1+\beta b}\, Ak^{\alpha} = \frac{\beta\alpha/(1-\alpha\beta)}{1/(1-\alpha\beta)}\, Ak^{\alpha} = \alpha\beta A k^{\alpha}.$$

This is identical to the policy $g(k) = \alpha\beta Ak^{\alpha}$ obtained in Problem 5, confirming that sequential and recursive methods yield the same solution.

Problem 7 – Recursive formulation: the periodic NGM

Consider a variant of the NGM in which total factor productivity alternates deterministically: $A_t = A_h$ in even periods and $A_t = A_l$ in odd periods ($A_h > A_l > 0$). Output is $A_t f(k_t)$ with $f(k) = k^{\alpha}$, $\delta = 1$, $u(c) = \log c$, and $0 < \beta < 1$.

1. Explain why the standard single-state Bellman equation $V(k) = \max\{u(c) + \beta V(k')\}$ does *not* represent this problem correctly. What additional information must the state include?

2. Write the problem as two linked Bellman equations $V_h(k)$ and $V_l(k)$.

3. Guess $V_h(k) = a_h + b\log k$ and $V_l(k) = a_l + b\log k$ (same slope b). Derive b and the policy functions $g_h(k)$ (capital chosen in a high-TFP period) and $g_l(k)$ (capital chosen in a low-TFP period).

4. Show that the saving rate $k'/(A_i k^{\alpha})$ is the same in both states and equals the constant $\alpha\beta$ from Problem 5. Provide an economic interpretation.

Solutions – Problem 7

Answer to 1. In the standard stationary Bellman equation, the agent faces the same problem at every date given the same capital stock. Here, the return on investment differs systematically between even and odd periods: in a high-productivity period the agent knows productivity will be A_l next period, whereas in a low-productivity period it will be A_h next period. The problem is *not* stationary if only k is the state – the optimal decision depends on whether the current period is even (high) or odd (low). The state must be augmented to (k, i) where $i \in \{h, l\}$ indicates the current TFP level.

Answer to 2. The linked Bellman equations are

$$V_h(k) = \max_{k'\geq 0}\big\{\log(A_h k^{\alpha} - k') + \beta V_l(k')\big\},$$
$$V_l(k) = \max_{k'\geq 0}\big\{\log(A_l k^{\alpha} - k') + \beta V_h(k')\big\}.$$

After a high-TFP period the continuation value is V_l (next period is low-TFP), and vice versa.

Answer to 3. Substitute $V_h(k) = a_h + b\log k$ and $V_l(k) = a_l + b\log k$ into the Bellman equations. From V_l, the first-order condition is

$$\frac{1}{A_l k^{\alpha} - k'} = \frac{\beta b}{k'} \implies k' = \frac{\beta b}{1+\beta b}\, A_l k^{\alpha} \equiv g_l(k).$$

The coefficient of $\log k$ on the right-hand side of the V_l equation is $\alpha(1+\beta b)$ (the same algebra as in Problem 6). Matching with b: $b = \alpha(1+\beta b)$, giving $b = \alpha/(1-\alpha\beta)$ – *the same value as in the non-periodic case.* Similarly from V_h we get $g_h(k) = \beta b A_h k^\alpha/(1+\beta b)$. Substituting b:

$$g_h(k) = \alpha\beta A_h k^\alpha, \qquad g_l(k) = \alpha\beta A_l k^\alpha.$$

*Answer to **4**.* The saving rates are $g_h(k)/(A_h k^\alpha) = \alpha\beta$ and $g_l(k)/(A_l k^\alpha) = \alpha\beta$ – *identical in both states and equal to the non-periodic saving rate.*

Interpretation. With log utility, the income and substitution effects of a change in the return to saving exactly offset each other (as established in Problem 3, $\sigma = 1$ implies EIS= 1). A higher TFP A_h raises current output and therefore current income and wealth (income effect, favouring more current consumption) by exactly the same proportion as it raises the return on capital (substitution effect, favouring more saving). The two effects cancel, so the *fraction* of output saved is unchanged.

Problem 8 – Linearisation of the policy function

Consider the infinite-horizon NGM with CRRA utility $u(c) = (c^{1-\sigma}-1)/(1-\sigma)$, Cobb-Douglas production $f(k) = k^\alpha$ ($A = 1$), depreciation rate $\delta > 0$, discount factor β, and capital-income share α.

1. Find the steady-state pair $(\bar{k}, \bar{c})$ from the Euler equation and resource constraint evaluated at $k_{t+1} = k_t = \bar{k}$.

2. Using the formula from the text, show that the coefficient Θ in the linearised policy function satisfies

$$\Theta = \frac{\beta^2(1-\alpha)}{\sigma}\left(\frac{1}{\beta} - 1 + \delta\right)\left(\frac{1}{\alpha}\left(\frac{1}{\beta} - 1 + \delta\right) - \delta\right).$$

3. The slope a_1 of the linearised policy function $g(k) \approx \bar{k}(1-a_1) + a_1 k$ around $\bar{k}$ satisfies a quadratic. State the formula (equation (4.30) in the text) and explain why only the root in $(0,1)$ corresponds to the stable policy function.

4. Compute $(\bar{k}, \bar{c})$, Θ, and a_1 numerically for $\alpha = 0.33$, $\beta = 0.99$, $\delta = 0.10$, and $\sigma = 2$.

Solutions – Problem 8

*Answer to **1**.* At the steady state the Euler equation gives $1 = \beta(f'(\bar{k}) + 1 - \delta)$, so $f'(\bar{k}) = 1/\beta - 1 + \delta$. With $f(k) = k^\alpha$, $f'(\bar{k}) = \alpha\bar{k}^{\alpha-1}$, thus

$$\bar{k} = \left(\frac{\alpha}{1/\beta - 1 + \delta}\right)^{1/(1-\alpha)}.$$

Steady-state consumption from the resource constraint ($k_{t+1} = k_t$): $\bar{c} = f(\bar{k}) - \delta\bar{k} = \bar{k}^\alpha - \delta\bar{k}$.

*Answer to **2**.* The general formula is $\Theta = \beta^2 u'(\bar{c}) f''(\bar{k})/u''(\bar{c})$. For CRRA: $u'(\bar{c}) = \bar{c}^{-\sigma}$ and $u''(\bar{c}) = -\sigma\bar{c}^{-\sigma-1}$, so $u'/u'' = -\bar{c}/\sigma$. For Cobb-Douglas: $f''(\bar{k}) = \alpha(\alpha-1)\bar{k}^{\alpha-2} = -(1-\alpha)\alpha\bar{k}^{\alpha-2}$. Therefore

$$\Theta = \beta^2 \cdot \frac{-\bar{c}}{\sigma} \cdot \left[-(1-\alpha)\alpha\bar{k}^{\alpha-2}\right] = \frac{\beta^2(1-\alpha)\alpha\bar{c}\bar{k}^{\alpha-2}}{\sigma}.$$

Using $\alpha\bar{k}^{\alpha-2} = f'(\bar{k})/\bar{k} = (1/\beta - 1 + \delta)/\bar{k}$ and $\bar{c}/\bar{k} = \bar{k}^{\alpha-1} - \delta = (1/\alpha)(1/\beta - 1 + \delta) - \delta$:

$$\Theta = \frac{\beta^2(1-\alpha)}{\sigma}\left(\frac{1}{\beta} - 1 + \delta\right)\left(\frac{1}{\alpha}\left(\frac{1}{\beta} - 1 + \delta\right) - \delta\right).$$

*Answer to **3**.* Differentiating the functional Euler equation at the steady state yields the quadratic $\beta a_1^2 - (1 + \beta + \Theta)a_1 + 1 = 0$ (with $\Theta > 0$). The two roots are

$$a_1 = \frac{(1+\beta+\Theta) \pm \sqrt{(1+\beta+\Theta)^2 - 4\beta}}{2\beta}.$$

One root satisfies $a_1 \in (0,1)$ (convergent dynamics; the policy function crosses the 45-degree line with a slope less than one at steady state) and the other is greater than one (divergent dynamics, explosive capital). Only the smaller

root is consistent with the Bellman equation's maximisation: the larger root corresponds to a saddle-path that moves away from steady state (not the maximum on the right-hand side of the Bellman equation). Hence

$$a_1 = \frac{(1+\beta+\Theta) - \sqrt{(1+\beta+\Theta)^2 - 4\beta}}{2\beta}.$$

*Answer to **4**.* With $\alpha = 0.33$, $\beta = 0.99$, $\delta = 0.10$, $\sigma = 2$:

$$\frac{1}{\beta} - 1 + \delta = \frac{1}{0.99} - 0.9 \approx 0.1101,$$

$$\bar{k} = \left(\frac{0.33}{0.1101}\right)^{1/0.67} \approx 5.15, \quad \bar{c} = 5.15^{0.33} - 0.1 \times 5.15 \approx 1.20.$$

$$\Theta = \frac{0.99^2 \times 0.67}{2} \times 0.1101 \times \left(\frac{0.1101}{0.33} - 0.10\right)$$
$$\approx 0.4901 \times 0.1101 \times 0.2337 \approx 0.0126.$$

$$1 + \beta + \Theta \approx 2.0026,$$

$$a_1 = \frac{2.0026 - \sqrt{2.0026^2 - 4 \times 0.99}}{2 \times 0.99} \approx \frac{2.0026 - \sqrt{4.0104 - 3.96}}{1.98} \approx \frac{2.0026 - \sqrt{0.0504}}{1.98}$$
$$\approx \frac{2.0026 - 0.2245}{1.98} \approx \frac{1.7781}{1.98} \approx \mathbf{0.898}.$$

A slope of $a_1 \approx 0.90$ means that after a 1 % deviation from $\bar{k}$, the next period's capital is about 0.9 % away from $\bar{k}$ – convergence is slow (roughly 6–7 periods to halve the gap).

Problem 9 – MATLAB: Value function iteration for the NGM

This problem asks you to solve the standard NGM numerically using value function iteration (VFI). Use the parameters $\alpha = 0.33$, $\beta = 0.99$, $\delta = 0.10$, $\sigma = 2$, $A = 1$ (same as Problem 8).

1. Compute the steady state $(\bar{k}, \bar{c})$ analytically and set up a capital grid with $n_k = 300$ equally-spaced points on $[0.5\bar{k},\ 1.5\bar{k}]$.

2. Write a MATLAB program implementing VFI. Pre-compute the full $n_k \times n_k$ matrix of feasible consumption values and the corresponding utility matrix. Iterate until the sup-norm distance between successive value functions is below 10^{-8}.

3. Plot (i) the converged value function $V(k)$ and (ii) the policy function $g(k)$ together with the 45-degree line, marking the steady state.

4. At the grid point closest to $\bar{k}$, compute the numerical slope $[g(k_{i+1}) - g(k_{i-1})]/(k_{i+1} - k_{i-1})$ and compare it to the analytical a_1 from Problem 8.

Solutions – Problem 9

*Answer to **1–4**.* The complete MATLAB code below addresses all four questions.

```
%% Problem 9: Value Function Iteration for the Neoclassical Growth Model
% Parameters
alpha = 0.33;    % capital share
beta  = 0.99;    % discount factor
delta = 0.10;    % depreciation rate
sigma = 2.00;    % CRRA curvature (sigma=1 => log utility)
A     = 1.00;    % TFP

%% Q1: Steady state
r_ss  = 1/beta - 1 + delta;            % net return at steady state
k_ss  = (alpha * A / r_ss)^(1/(1-alpha));
```

```
c_ss  = A * k_ss^alpha - delta * k_ss;
fprintf('Steady state: k_ss = %.6f,  c_ss = %.6f\n', k_ss, c_ss);

% Capital grid
n_k    = 300;
k_min  = 0.5 * k_ss;
k_max  = 1.5 * k_ss;
k_grid = linspace(k_min, k_max, n_k)';   % (n_k x 1) column vector

%% Q2: Pre-compute consumption and utility matrices
% y_grid(i) = resources available when k = k_grid(i)
y_grid = A * k_grid.^alpha + (1 - delta) * k_grid;   % (n_k x 1)

% C_mat(i,j) = consumption when current capital is k_grid(i)
%              and next-period capital is k_grid(j)
C_mat = bsxfun(@minus, y_grid, k_grid');   % (n_k x n_k), broadcasts

% Set infeasible entries to -Inf in utility matrix
if sigma == 1
    U_mat = log(C_mat);
else
    U_mat = (C_mat.^(1-sigma) - 1) ./ (1 - sigma);
end
U_mat(C_mat <= 0) = -Inf;   % infeasible choices

% VFI loop
V        = zeros(n_k, 1);   % initial guess: V_0 = 0
tol      = 1e-8;
diff     = 1;
iter     = 0;
max_iter = 3000;

while diff > tol && iter < max_iter
    iter = iter + 1;

    % Bellman equation: maximise over k' (columns) for each k (rows)
    RHS = U_mat + beta * repmat(V', n_k, 1);   % (n_k x n_k)
    [V_new, k_idx] = max(RHS, [], 2);           % max over columns

    diff = max(abs(V_new - V));
    V    = V_new;
end

fprintf('Converged in %d iterations (sup-norm = %.2e)\n', iter, diff);

% Policy function for capital and consumption
k_pol = k_grid(k_idx);
c_pol = y_grid - k_pol;

%% Q3: Plots
figure('Position', [100 100 1000 420]);

subplot(1, 2, 1)
plot(k_grid, V, 'b-', 'LineWidth', 1.8)
xline(k_ss, 'r--', '\bar{k}', 'LabelVerticalAlignment', 'bottom', ...
    'FontSize', 11)
xlabel('Capital k',     'FontSize', 12)
ylabel('V(k)',          'FontSize', 12)
title('Value Function', 'FontSize', 13)
grid on

subplot(1, 2, 2)
plot(k_grid, k_pol, 'b-', 'LineWidth', 1.8)
hold on
```

```
plot(k_grid, k_grid, 'k--', 'LineWidth', 1.0)
xline(k_ss, 'r--', '\bar{k}', 'LabelVerticalAlignment', 'bottom', ...
    'FontSize', 11)
xlabel('Current capital k',       'FontSize', 12)
ylabel('Next-period capital k''', 'FontSize', 12)
title('Policy Function g(k)',     'FontSize', 13)
legend('g(k)', '45^o line', 'Location', 'northwest', 'FontSize', 11)
grid on

%% Q4: Numerical slope at steady state vs analytical a_1
% Find grid index closest to k_ss
[~, i_ss] = min(abs(k_grid - k_ss));
i_lo = max(i_ss - 1, 1);
i_hi = min(i_ss + 1, n_k);

slope_num = (k_pol(i_hi) - k_pol(i_lo)) / (k_grid(i_hi) - k_grid(i_lo));

% Analytical a_1 from Problem 8
Theta = beta^2 * (1-alpha) / sigma * r_ss * (r_ss/alpha - delta);
a1    = (1 + beta + Theta - sqrt((1 + beta + Theta)^2 - 4*beta)) / (2*beta);

fprintf('Numerical slope at k_ss:  %.6f\n', slope_num);
fprintf('Analytical a_1 (Prob. 8): %.6f\n', a1);
```

Listing 1: VFI for the NGM – Problem 9

Expected output (approximate):

```
Steady state: k_ss = 5.152063,  c_ss = 1.202044
Converged in 183 iterations (sup-norm = 9.96e-09)
Numerical slope at k_ss:  0.897
Analytical a_1 (Prob. 8): 0.898
```

The close agreement between the numerical slope and $a_1 \approx 0.898$ from Problem 8 confirms that the linearisation is an accurate local approximation.

Problem 10 – MATLAB: Finite-horizon saving rates and capital convergence

Return to the log/Cobb-Douglas/full-depreciation case: $u(c) = \log c$, $f(k) = Ak^\alpha$, $\delta = 1$, $A = 1$, $\alpha = 0.33$, $\beta = 0.99$. The analytical saving rate in a T-period economy is (from the text)

$$s_t(T) = \alpha\beta \, \frac{1 - (\alpha\beta)^{T-t}}{1 - (\alpha\beta)^{T-t+1}}, \qquad t = 0, 1, \ldots, T-1.$$

The infinite-horizon saving rate is $s^\infty = \alpha\beta$.

1. Write a MATLAB function `saving_rates(T,alpha,beta)` that returns the vector $(s_0, s_1, \ldots, s_{T-1})$ using the formula above.

2. Plot the saving-rate profiles for $T \in \{5, 10, 20, 50\}$ on a single figure, together with the infinite-horizon rate s^∞ as a horizontal dashed line.

3. Simulate the capital path $\{k_t\}$ starting from $k_0 = 0.5\bar{k}$ (where $\bar{k} = (\alpha\beta)^{1/(1-\alpha)}$) for $T \in \{5, 10, 20, 50, \infty\}$ and plot all paths together.

4. For the infinite-horizon case, compute the *half-life* of convergence: the number of periods τ^* such that $|k_{\tau^*} - \bar{k}| \le \frac{1}{2}|k_0 - \bar{k}|$. Comment on how it relates to the slope α of the policy function.

Solutions – Problem 10

*Answer to **1–4**.* The complete MATLAB code below addresses all four questions.

```
%% Problem 10: Finite-horizon saving rates and capital paths
alpha  = 0.33;
beta   = 0.99;
A      = 1;        % TFP (with full depreciation)
s_inf  = alpha * beta;                          % infinite-horizon saving rate
k_ss   = (alpha * beta * A)^(1/(1-alpha));      % steady state

%% Q1: Function to compute saving rates for a T-period economy
saving_rates = @(T) arrayfun(@(t) ...
    alpha*beta*(1-(alpha*beta)^(T-t)) / (1-(alpha*beta)^(T-t+1)), ...
    0:T-1);

%% Q2: Plot saving-rate profiles
T_vals = [5, 10, 20, 50];
clrs   = {'b','r','m','g'};

figure('Position', [100 100 1100 450]);
subplot(1, 2, 1)
hold on
for j = 1:numel(T_vals)
    T   = T_vals(j);
    s_t = saving_rates(T);
    plot(0:T-1, s_t, [clrs{j} '-o'], 'MarkerSize', 4, 'LineWidth', 1.4, ...
        'DisplayName', ['T = ' num2str(T)]);
end
yline(s_inf, 'k--', 'LineWidth', 1.5, 'DisplayName', 'T = \infty');
xlabel('Period t',       'FontSize', 12)
ylabel('Saving rate s_t','FontSize', 12)
title('Optimal Saving Rates','FontSize', 13)
legend('Location', 'southeast', 'FontSize', 10)
grid on
xlim([-0.5, 50])

%% Q3: Simulate capital paths from k_0 = 0.5 * k_ss
N_sim = 55;   % simulation length
subplot(1, 2, 2)
hold on

for j = 1:numel(T_vals)
    T     = T_vals(j);
    s_t   = saving_rates(T);
    T_sim = min(T, N_sim);
    k     = zeros(1, T_sim + 1);
    k(1)  = 0.5 * k_ss;
    for t = 1:T_sim
        k(t+1) = s_t(t) * A * k(t)^alpha;
    end
    plot(0:T_sim, k, [clrs{j} '-'], 'LineWidth', 1.4, ...
        'DisplayName', ['T = ' num2str(T)]);
end

% Infinite-horizon path: k_{t+1} = s_inf * k_t^alpha
k_inf    = zeros(1, N_sim + 1);
k_inf(1) = 0.5 * k_ss;
for t = 1:N_sim
    k_inf(t+1) = s_inf * A * k_inf(t)^alpha;
end
plot(0:N_sim, k_inf, 'k--', 'LineWidth', 1.5, 'DisplayName', 'T = \infty');
yline(k_ss, 'k:', '\bar{k}', 'LabelHorizontalAlignment','left', 'FontSize',11);

xlabel('Period t',                      'FontSize', 12)
ylabel('Capital k_t',                   'FontSize', 12)
title('Capital Path from k_0 = 0.5\bar{k}', 'FontSize', 13)
legend('Location', 'southeast', 'FontSize', 10)
```

```
grid on

%% Q4: Half-life of convergence (infinite-horizon)
gap      = abs(k_inf - k_ss);
half_gap = gap(1) / 2;
tau_star = find(gap <= half_gap, 1) - 1;   % first period gap <= half initial gap
fprintf('Half-life of convergence: %d periods\n', tau_star);

% Analytical approximation: the linearised map gives k_{t+1} - k_ss ~ alpha*(k_t - k_ss)
% => half-life ~ log(0.5)/log(alpha)
tau_analytic = log(0.5) / log(alpha);
fprintf('Analytical half-life (log-linearisation): %.2f periods\n', tau_analytic);
```

Listing 2: Finite vs. infinite horizon – Problem 10

Interpretation of Question 4. The policy function for the infinite-horizon log-CD-full-depreciation economy is $g(k) = \alpha\beta A k^{\alpha}$. Its derivative at $\bar{k}$ is

$$g'(\bar{k}) = \alpha^2 \beta A \bar{k}^{\alpha-1} = \alpha \times (\alpha\beta A \bar{k}^{\alpha-1}) = \alpha,$$

since $\alpha\beta A\bar{k}^{\alpha-1} = 1$ at the steady state (from $\bar{k} = (\alpha\beta A)^{1/(1-\alpha)}$). Therefore, locally around $\bar{k}$:

$$k_{t+1} - \bar{k} \approx \alpha(k_t - \bar{k}),$$

and the gap halves every $\tau^* = \log(1/2)/\log(1/\alpha) = \log 2/\log(1/\alpha)$ periods. For $\alpha = 0.33$: $\tau^* = \log 2/\log 3 \approx 0.63$ periods – the economy is essentially at steady state in one or two periods. This rapid convergence is a feature of full depreciation ($\delta = 1$); with partial depreciation the policy function has a slope closer to 1 and convergence is much slower (as seen in Problem 8).